unnecessary latency in our application, reducing the advantage of using a cache and possibly even making it counterproductive.

7.7.6 Other Approaches to Concurrency

Using locking mechanisms is not the only way to deal with concurrency.

Locking handles the access to segment of code that mutates a resource, so that we are sure that these instructions applying modification will be executed by one thread, and one thread only, at each instant in time.

On the other end of the spectrum, there is a completely symmetrical approach, championed by functional programming: remove altogether mutability from your resources.

One of the principles of functional programming is, in fact, that all objects should be immutable. This has several advantages, for instance it makes it easier to reason about code because we can analyze a function being sure that no other method will influence what it does, and it removes race conditions altogether. It also has some disadvantages, for instance it makes harder some operations, like keeping a running state, and make it more expensive updating state in large objects.

When we talk about concurrency, having immutable objects means that you can read your resource without worrying that